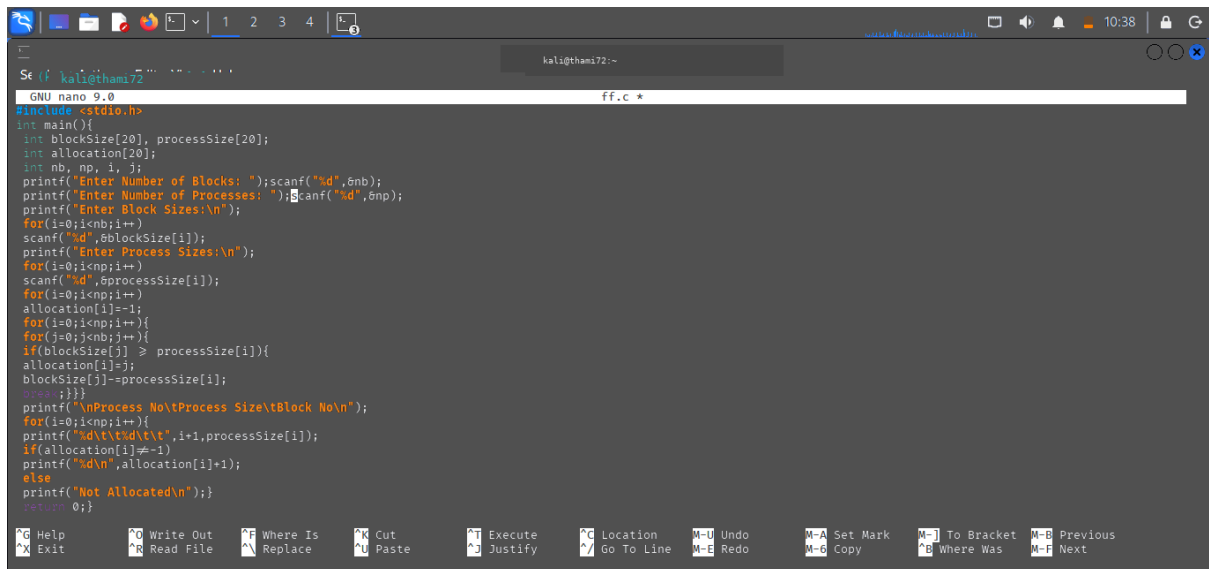
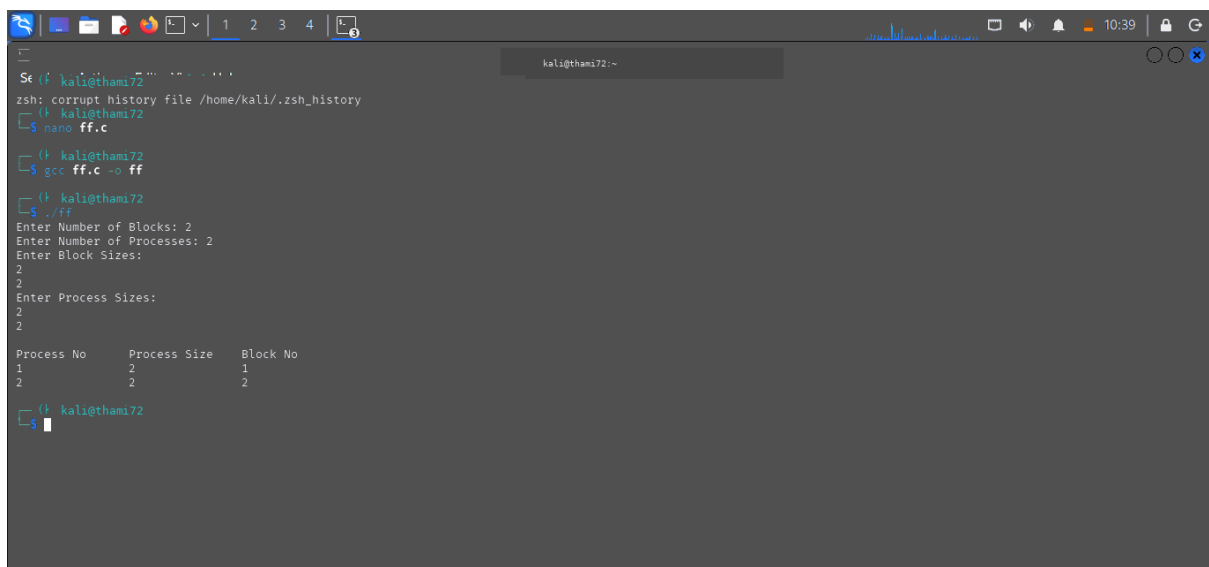


C PROGRAMMING :FIRST FIT



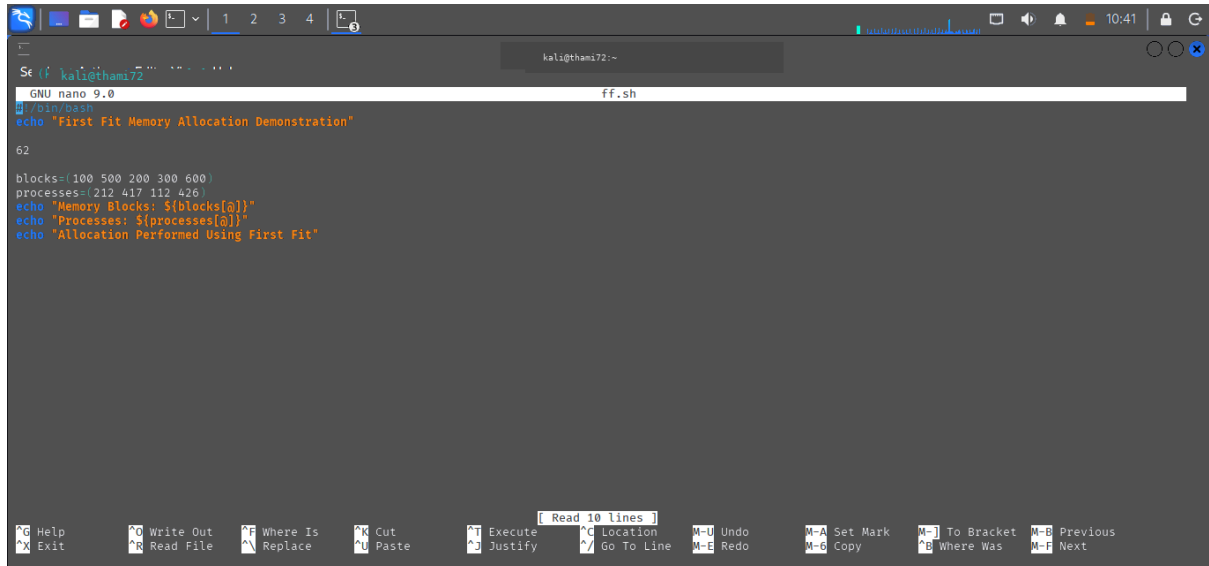
```
GNU nano 9.0 ff.c *
#include <stdio.h>
int main(){
    int blockSize[20], processSize[20];
    int allocation[20];
    int nb, np, i, j;
    printf("Enter Number of Blocks: ");scanf("%d",&nb);
    printf("Enter Number of Processes: ");scanf("%d",&np);
    printf("Enter Block Sizes:\n");
    for(i=0;i<nb;i++){
        scanf("%d",&blockSize[i]);
        printf("Enter Process Sizes:\n");
        for(i=0;i<np;i++){
            scanf("%d",&processSize[i]);
            for(j=0;j<nb;j++){
                allocation[j]=-1;
                if(blockSize[j] >= processSize[i]){
                    allocation[j]=i;
                    blockSize[j]-=processSize[i];
                    break;
                }
            }
            printf("\nProcess No\tProcess Size\tBlock No\n");
            for(i=0;i<np;i++){
                printf("%d\t%d\t\t",i+1,processSize[i]);
                if(allocation[i]!=-1)
                    printf("%d\n",allocation[i]+1);
                else
                    printf("Not Allocated\n");
            }
        }
    }
    return 0;
}
```

OUTPUT :



```
kali@thami72:~$ gcc ff.c -o ff
kali@thami72:~$ ./ff
Enter Number of Blocks: 2
Enter Number of Processes: 2
Enter Block Sizes:
2
2
Enter Process Sizes:
2
2
Process No      Process Size    Block No
1                2                1
2                2                2
kali@thami72:~$
```

SHELL PROGRAMMINNG : FIRST FIT

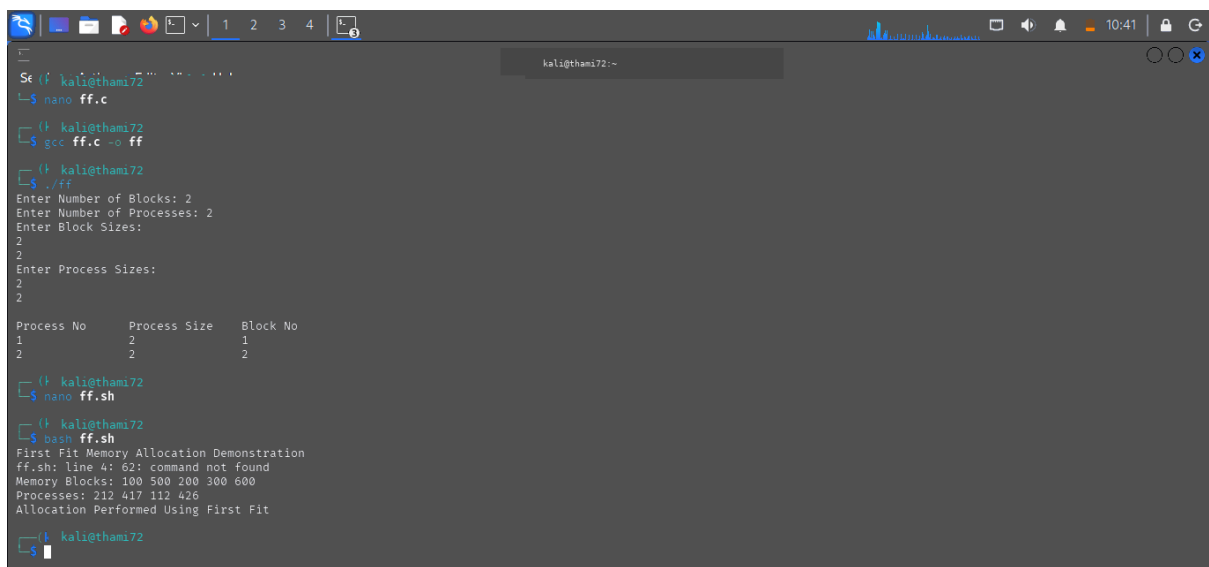


```
GNU nano 9.0 ff.sh
#!/bin/bash
echo "First Fit Memory Allocation Demonstration"

62

blocks=(100 500 200 300 600)
processes=(212 417 112 426)
echo "Memory Blocks: ${blocks[@]}"
echo "Processes: ${processes[@]}"
echo "Allocation Performed Using First Fit"
```

OUTPUT :



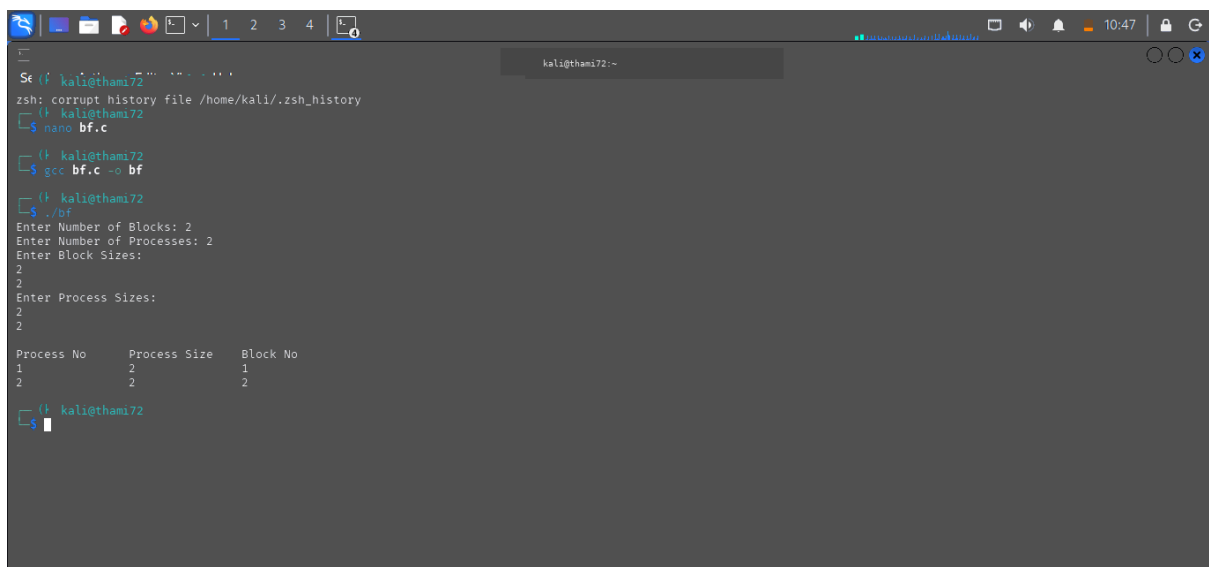
```
Se (f kali@thami72 ~)
kali@thami72 ~
$ nano ff.c
kali@thami72 ~
$ gcc ff.c -o ff
kali@thami72 ~
$ ./ff
Enter Number of Blocks: 2
Enter Number of Processes: 2
Enter Block Sizes:
2
2
Enter Process Sizes:
2
2
Process No      Process Size    Block No
1                2                1
2                2                2
kali@thami72 ~
$ nano ff.sh
kali@thami72 ~
$ bash ff.sh
First Fit Memory Allocation Demonstration
ff.sh: line 4: 62: command not found
Memory Blocks: 100 500 200 300 600
Processes: 212 417 112 426
Allocation Performed Using First Fit
kali@thami72 ~
$
```

C PROGRAMMING : BEST FIT



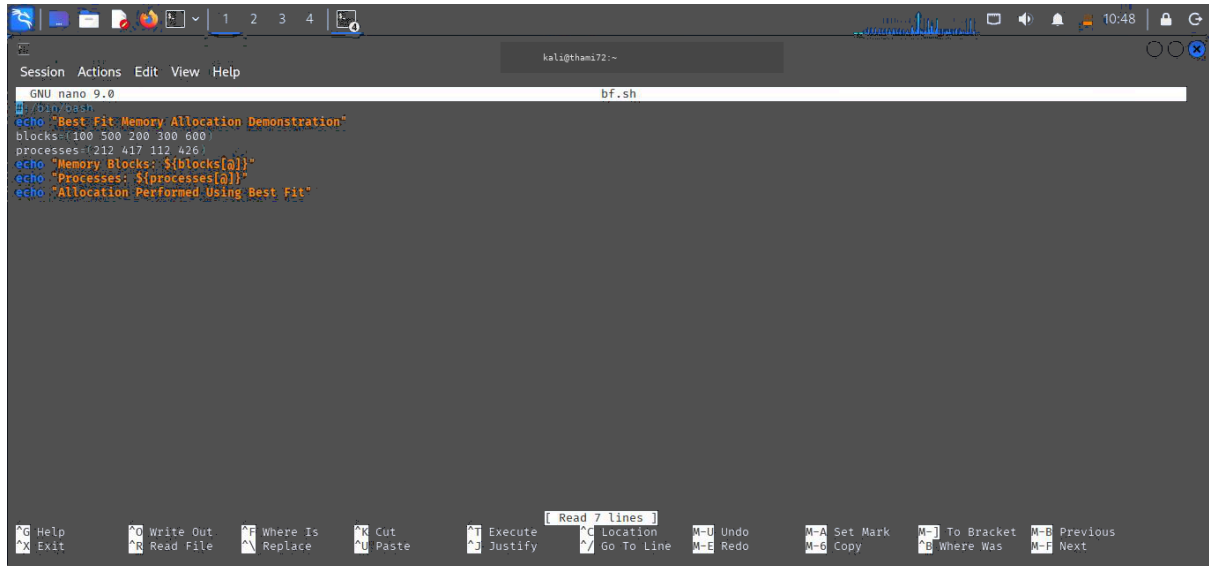
```
GNU nano 9.0 bf.c *
#include <stdio.h>
int main()
{
    int blockSize[20], processSize[20];
    int allocation[20];
    int nb,np,i,j,bestIdx;
    printf("Enter Number of Blocks: ");scanf("%d",&nb);
    printf("Enter Number of Processes: ");scanf("%d",&np);
    printf("Enter Block Sizes:\n");
    for(i=0;i<nb;i++)
        scanf("%d",&blockSize[i]);printf("Enter Process Sizes:\n");
    for(i=0;i<np;i++)
        scanf("%d",&processSize[i]);
    for(i=0;i<np;i++){
        for(j=0;j<nb;j++){
            if(blockSize[j] >= processSize[i]){
                if(bestIdx==-1 || blockSize[j] < blockSize[bestIdx])bestIdx=j;
            }
        }
        if(bestIdx!=-1){
            allocation[i]=bestIdx;
            blockSize[bestIdx]=processSize[i];
            printf("\nProcess No\tProcess Size\tBlock No\n");
            for(i=0;i<np;i++){
                printf("%d\t%d\t%d",i+1,processSize[i],allocation[i]);
                if(allocation[i]!=-1)
                    printf("%d\n",allocation[i]+1);
                else
                    printf("Not Allocated\n");
            }
        }
    }
    return 0;
}
```

OUTPUT :



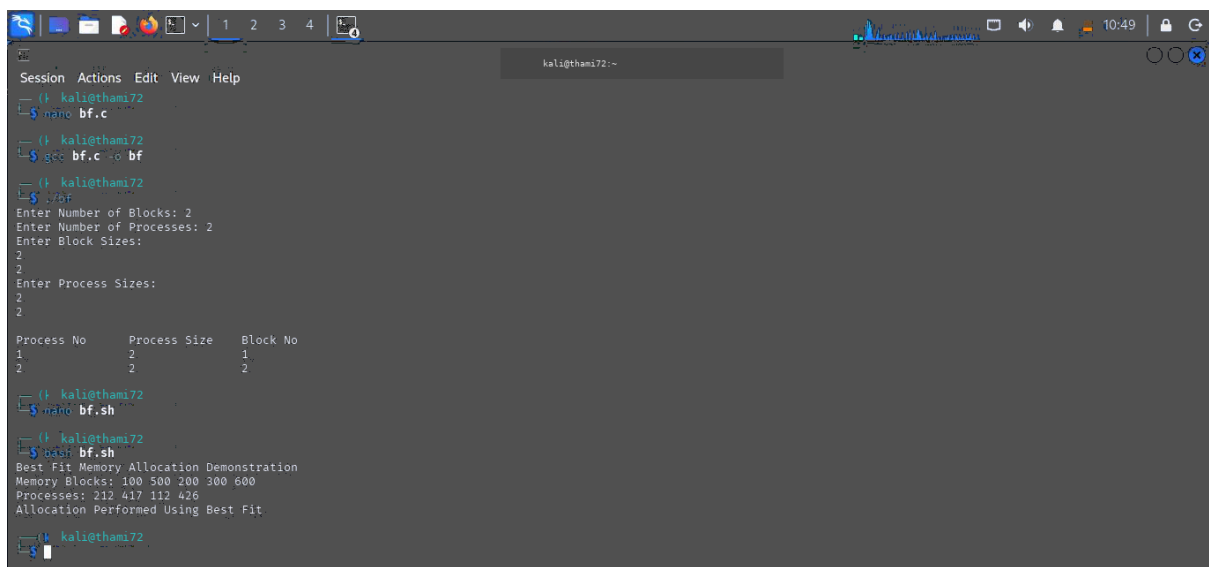
```
kali@thami72:~
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72:~
$ nano bf.c
kali@thami72:~
$ gcc bf.c -o bf
kali@thami72:~
$ ./bf
Enter Number of Blocks: 2
Enter Number of Processes: 2
Enter Block Sizes:
2
2
Enter Process Sizes:
2
2
Process No      Process Size    Block No
1               2               1
2               2               2
kali@thami72:~
$
```

SHELL PROGRAMMING : BEST FIT



```
GNU nano 9.0 bf.sh
#!/bin/bash
echo "Best Fit Memory Allocation Demonstration"
blocks=(100 500 200 300 600)
processes=(212 417 112 426)
echo "Memory Blocks: ${blocks[@]}"
echo "Processes: ${processes[@]}"
echo "Allocation Performed Using Best Fit"
```

OUTPUT :



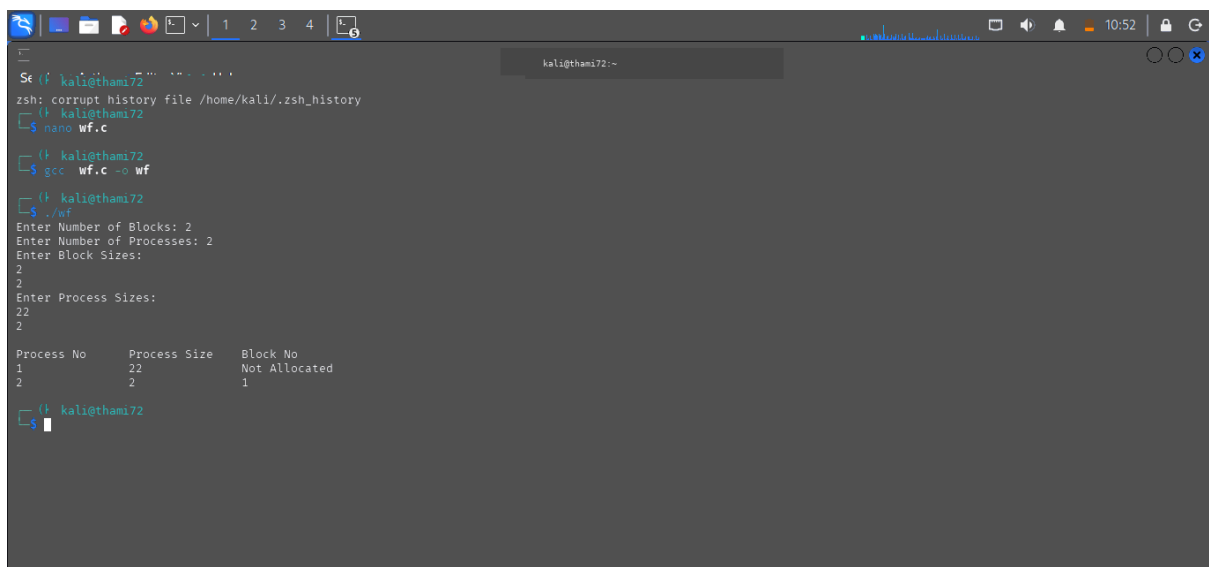
```
kali@thami72:~$ nano bf.c
kali@thami72:~$ gcc bf.c -o bf
kali@thami72:~$ ./bf
Enter Number of Blocks: 2
Enter Number of Processes: 2
Enter Block Sizes:
2
2
Enter Process Sizes:
2
2
Process No      Process Size    Block No
1                2                1
2                2                2
kali@thami72:~$ nano bf.sh
kali@thami72:~$ ./bf.sh
Best Fit Memory Allocation Demonstration
Memory Blocks: 100 500 200 300 600
Processes: 212 417 112 426
Allocation Performed Using Best Fit
kali@thami72:~$
```

C PROGRAMMING : WORST FIT



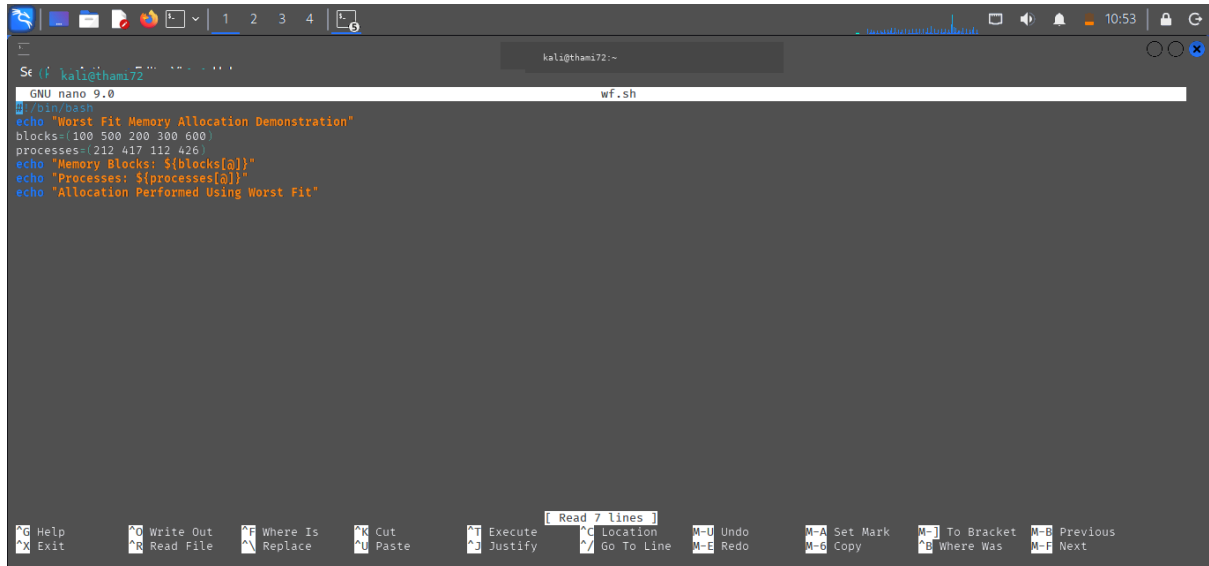
```
GNU nano 9.0 wf.c *
#include <stdio.h>
int main(){
    int blockSize[20], processSize[20];
    int allocation[20];
    int nb,np,i,j,worstIdx;
    printf("Enter Number of Blocks: ");scanf("%d",&nb);
    printf("Enter Number of Processes: ");scanf("%d",&np);
    printf("Enter Block Sizes:\n");
    for(i=0;icnb;i++){
        scanf("%d",&blockSize[i]);printf("Enter Process Sizes:\n");
        for(j=0;jnp;j++){
            scanf("%d",&processSize[j]);
            for(i=0;icnp;i++){
                allocation[i]=-1;
                for(i=0;icnp;i++){worstIdx=-1;
                    for(j=0;jcnb;j++){
                        if(blockSize[j] >= processSize[i]){
                            if(worstIdx== -1 || blockSize[j] > blockSize[worstIdx])worstIdx=j;}}
                    if(worstIdx!= -1){
                        allocation[i]=worstIdx;
                        blockSize[worstIdx]=processSize[i];}}
                printf("\nProcess No\tProcess Size\tBlock No\n");
                for(i=0;icnp;i++){
                    printf("%d\t\t%d\t\t",i+1,processSize[i]);
                    if(allocation[i]!=-1)
                        printf("%d\n",allocation[i]+1);
                    else
                        printf("Not Allocated\n");}
                return 0;}
```

OUTPUT :



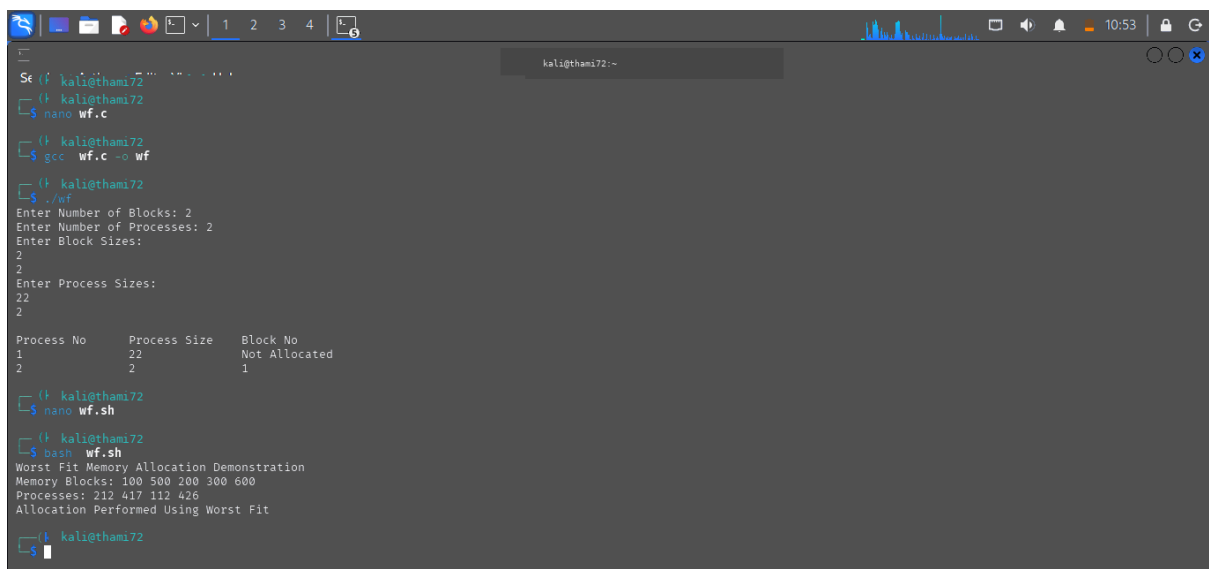
```
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72 ~
$ nano wf.c
kali@thami72 ~
$ gcc wf.c -o wf
kali@thami72 ~
$ ./wf
Enter Number of Blocks: 2
Enter Number of Processes: 2
Enter Block Sizes:
2
2
Enter Process Sizes:
22
2
Process No      Process Size    Block No
1                22              Not Allocated
2                 2                1
```

SHELL PROGRAMMING : WORST FIT



```
GNU nano 9.0 wf.sh
#!/bin/bash
echo "Worst Fit Memory Allocation Demonstration"
blocks=(100 500 200 300 600)
processes=(212 417 112 426)
echo "Memory Blocks: ${blocks[@]}"
echo "Processes: ${processes[@]}"
echo "Allocation Performed Using Worst Fit"
```

OUTPUT :



```
kali@thami72:~$ nano wf.c
kali@thami72:~$ gcc wf.c -o wf
kali@thami72:~$ ./wf
Enter Number of Blocks: 2
Enter Number of Processes: 2
Enter Block Sizes:
2
2
Enter Process Sizes:
22
2
Process No      Process Size    Block No
1                22              Not Allocated
2                2               1
kali@thami72:~$ nano wf.sh
kali@thami72:~$ bash wf.sh
Worst Fit Memory Allocation Demonstration
Memory Blocks: 100 500 200 300 600
Processes: 212 417 112 426
Allocation Performed Using Worst Fit
kali@thami72:~$
```